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Original article

Cross sectional evaluation of communication skills attitude in undergraduate medical students

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Abstract

Good communication skills have become a useful asset in building an honest doctor patient relationship. It has become an indigenous part of comprehensive medical care. Cross-sectional observational study was conducted after Institutional Ethics Committee permission. Demographic information and CSAS was administered to students and collected after 20-30 minutes. Total 400 undergraduate medical students, 100 students from each of first, second, third and fourth year of MBBS course were included. CSAS consists of 26 items, of which 13 were positive and 13 were negative statements. In CSAS scale, mean PAS was 55.53±11.12 and mean NAS was 30.55±13.66. In female medical students PAS was 52.96±10.12 statistically significant (p=0.013) compared to male students PAS of 50.21± 11.89. In female medical students NAS was 29.64±12.80 statistically significant (p=0.034) compared to males NAS of 32.24±11.57. In medical students with doctors parents PAS was 50.32±11.30 statistically significant (p=0.026) compared to medical students with non-doctors parents was 53.08±11.44. The PAS for students from rural address was 52.17±10.33 that was statistically significant (p=0.027) compared to PAS for Urban address 50.14±10.34. The average self rating score for communication skills was 6.06±1.52. 92% students feel need to improve their communication skills and 55.52% preferred role play method for learning communication skills. There is destitution in positive attitude and negative attitude of communication skills in medical undergraduates. Communication skills need to be assessed in undergraduate medical students in India.

Key words: Competency, Negative attitude, Positive attitude, Skill

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World Health Organization (WHO) has specified qualities that are important for a physician this includes that of a caregiver, who knows community health requirements, reconciles individual, encourage healthy lifestyles, uses of new technologies, and has good communi-

cation to be able to work efficiently in teams[1,2,3]. A practicing physician comes across approximately 160,000 to 300,000 encounters with patients during his or her lifetime career where CS is considered decisive[4,5].

In countries where the communication gap exists between the doctor and patient, effective communication doesn't happen[6]. Communication skills are important for physicians to elicit relevant history that will be helpful for precise diagnosis and also convincing the patient to follow medical advice[2,7].

The new competency based curriculum includes Foundation course which incorporates an orientation of undergraduate students to communication skills[8]. This study was planned to understand the opinion of the medical students towards the need of learning and developing communication skills as a core competency for becoming an efficient healthcare provider.

Material and Methods

Institutional Ethics Committee (IEC) permission was obtained before initiating this study. It was a questionnaire based cross sectional observational study on assessment of communication skills among the undergraduate medical students. Informed written consent was administered and obtained prior to enrollment of each student in the study. The inclusion criteria were age more than 18 yrs of age; both gender and willing to sign the written inform consent form. After explaining the purpose of the study to the students, these students were requested to provide the demographic information in predesigned record form. To evaluate the attitude of medical students towards communication skills Communication Skills Attitude Scale (CSAS) containing 30 items were given to students and collected after a period of 40 to 50 minutes.

Those who were not willing to participate, or did not return or returned incompletely filled forms were excluded from the study. In this study CSAS was distributed to 464 students that had participated out of which 418 returned the completely filled questionnaire. Depending on inclusion and exclusion criteria, 100 students from each of first, second, third and fourth year, total 400 undergraduate medical students of medical college and tertiary hospital were selected.

The demographic profile contained information on self-rating of CS and the importance of CS in medical practice on the scale of 1 to10 where 01 being lowest and 10 being highest. CSAS consists of 26 items as shown given in Table 2 of which 13 were positive statements and 13 were negative statements. All the items in CSAS attitude scale were graded from 1 (strongly disagree) to 5 (strongly agree) on 5-point Likert scale. The 30 item questionnaire consist of positive and negative items, the negative attitude scale (NAS) score was obtained by adding the scores of items 2, 3, 6, 8, 11, 13, 15, 18, 19, 20, 22, 24, 26. The remaining items are positive attitude scale (PAS) score obtained by adding the scores of items 1, 4, 5, 7, 9, 10, 12, 14, 16, 17, 21, 23, 25. Both, PAS and NAS range from 13 to 65[6,9].

The CSAS was validated by five experts from the medical education department for content and construct validity. After receiving the comments from experts, demographic profile was modified according to local needs. The CSAS was pretested by administering it to 10 medical undergraduates and again after 7 days for reliability. The reliability coefficient Cronbach's alpha was 0.80 for PAS and 0.72 for NAS for the subscales of CSAS.

Results

In our study 316 (79%) students were from urban area and 84 (21%) were from rural area. There were 165 (41.25%) students from English medium of education while 235 (49.75%) were from Vernacular medium. There were 197 (49.25%) males and 203 (50.75%) females. 368 (92%) students felt the need to improve communication skills and 303 (75.75%) responded by stating that they would like to attend sessions on communication skills. There were 310 (77.5%) students that would like to attend session on Communication Skills if conducted without being assessed. The age, self scoring and importance of CS in medical practice is mentioned in table 1.

With relation to importance of communication skills in medicine 239 (59.75 %) felt that CS is very important, 154 (38.5%) responded as important, 6 (1.5%) responded as slightly important while no student felt that CS is not important or not important at all. 45 (11.25%) students had attended classes on personality development. The mean CSAS score of each item is given in the table 2.

Considering parents education, there were 19 students and their parents were less than 10 standard, 45 students had parents education between 10-12 standard, 83 students parents were 12th pass, 141 students parents were graduates while 112 students parents were post graduates. With regard to parents profession there were 125 (31.25%) students whose parents were doctors and 275 (68.75%) students parents were not from medical profession.

Table 1: Age, self rating and importance of communication skills in medical practice by students

Items	I MBBS	II MBBS	III MBBS	IV MBBS	Average
Age (years)	18.95 ± 0.73	19.62 ± 0.95	20.45 ± 1.23	21.57 ± 2.14	20.14 ± 1.26
Self rating of CS	6.52 ± 1.55	5.79 ±1.36**	6.03 ± 1.64	5.93 ± 1.53#	6.06 ± 1.52
Importance of CS in medical practice	9.18 ± 1.07	8.85 ± 1.48	8.89 ± 1.27	9.03 ± 1.42	8.98 ± 1.31

*I MBBS versus II MBBS, # I MBBS versus IV MBBS [**p<0.01, # p<0.05]

Table 2: Assessment of scores of Communication Skills Attitude Scale (CSAS) items

Sr. No	Items	Subscale	Mean ± SD
1	In order to be a good doctor I must have good communication skills	PAS	4.44 ± 0.11
2	I don't see why I should learn communication skills	NAS	1.99 ± 0.14
3	Nobody is going to fail in medical degree for having poor communication skills	NAS	2.56 ± 0.28
4	Developing my communication skills is just as important as developing my knowledge of Medicine	PAS	4.21± 0.11
5	Learning communication skills will help me respect patients	PAS	4.24 ± 0.13
6	I haven't got time to learn communication skills	NAS	2.36 ± 0.21
7	Learning communication skills is interesting	PAS	3.87 ± 0.21
8	It is too much trouble to attend sessions on communication skills	NAS	2.31 ± 0.31
9	Learning communication skills will help improve my team-working skills	PAS	4.24 ± 0.13
10	Learning communication skills will improve my ability to communicate with patients	PAS	4.30 ± 0.14
11	Communication skills teaching states the obvious and then complicates it	NAS	2.59 ± 0.19
12	Learning communication skills is fun	PAS	3.57 ± 0.18
13	Learning communication skills is too easy	NAS	2.48 ± 0.25
14	Learning communication skills will help me respect my colleagues	PAS	3.98 ± 0.14
15	I find it difficult to trust information about communication skills given to me by non-clinical lecturers	NAS	2.53 ± 0.29
16	Learning communication skills will help me recognize patient's rights regarding confidentiality and informed consent	PAS	3.99 ± 0.15
17	Communication skills teaching would have a better image if it sounded more like a science subject	PAS	3.24 ± 0.36
18	When applying for medicine, I thought it was a really good idea to learn communication skills	NAS	2.51 ± 0.33
19	I dont need good communication skills to be a doctor	NAS	1.96 ± 0.02
20	I dont want to tell anyone that I am having problems with my communica-tion skills	NAS	2.53 ± 0.20
21	I think its really useful learning communication skills in medical degree	PAS	4.01 ± 0.17
22	My ability to pass exams will get me through medical school rather than my ability to communicate	NAS	2.32 ± 0.19
23	Learning communication skills is applicable to learning medicine	PAS	3.85 ± 0.15
24	I find it difficult to take communication skills learning seriously	NAS	2.37 ± 0.18
25	Learning communication skills is important because of my ability to communicate with patients for lifelong	PAS	4.01 ± 0.15
26	Communication skills learning should be left to psychology students, not medical students	NAS	2.06 ± 0.06

PAS: Positive Attitude Scale, NAS: Negative Attitude Scale

Table 3: Comparison of CSAS in different academic years

Items	I MBBS	II MBBS	III MBBS	IV MBBS	Average
PAS	58.61± 11.86*	56.96±11.19	54.95 ± 11.97	51.69 ± 10.97	55.53 ± 11.12
NAS	28.87 ± 15.48	29.61 ± 12.96	31.18 ± 13.11	32.52 ± 13.07	30.55 ± 13.66

* I MBBS versus IV MBBS, [* p<0.05]

Table 4: Comparison of CSAS with respect to gender, address and parents profession

Comparison of CSAS according to Gender						
	Male		**Female**		**t/df**	**P**
	Mean	**SD**	**Mean**	**SD**		
PAS	50.21	11.89	52.96	10.12	2.49 (384)	0.013
NAS	32.24	11.57	29.64	12.80	0.45 (392)	0.034
Comparison of CSAS according to address						
	Urban		**Rural**		**t/df**	**P**
	Mean	**SD**	**Mean**	**SD**		
PAS	50.14	10.34	52.17	10.33	2.23 (130)	0.027
NAS	31.47	8.12	30.29	9.23	1.14 (145)	0.252
Comparison of CSAS according to Parents' profession						
	Parents are Doctors		**Parents are Non-Doctors**		**t/df**	**P**
	Mean	**SD**	**Mean**	**SD**		
PAS	50.32	11.30	53.08	11.44	2.24 (237)	0.026
NAS	31.07	12.11	30.40	09.21	0.55 (191)	0.582

Comparison of CSAS in different academic years is given in table 3. Comparison of CSAS according to gender, address and profession of parents is mentioned in table 4. For learning CS, 181 (55.52%) suggested role play, 82 (25.15%) suggested Case based problems, 50 (15.34%) suggested real/simulated patients and 13 (3.99%) suggested didactic lectures. Among 400 students, 326 (81.5%) opined that the knowledge of Communication Skills needs to be assessed.

Discussion

Physician's success depends on gaining patients confidence by good communication skills, this often increases patients' compliance and subsequently achieves better therapeutic response further strengthening the doctor–patient relationship. Attitude is one of the important determining factors for changing behavior in the teaching and learning process[10].

CSAS was developed by Rees and is used by many researchers across the world over as data collection tool for determining students attitude towards CS. CSAS is coherent, firm and dependable measure to learn medical student's attitudes towards communication skills [9,11,12,13].

In our study, PAS was prominent in first year and thereafter decreased in second, third and fourth year while NAS was lower in first year and thereafter increase in second, third and final year. In study by Shankar NAS scores were significantly lower among the third and fourth semester respondents[13]. Similar studies have shown that first and second year students scored significantly higher on the PAS than fourth year students[6,14,15,16]. In our study the decrease in PAS and increase in NAS might be due to students contact with patients during clinical posting in which they begin to understand the importance of effective communication. This decrease in PAS and increase in NAS need to be halted by the effective communication skills module from first year of medical curriculum.

In study by Lumma-Sellenthin female students had more positive attitude towards communication skills training than their male peers[17]. Similarly other studies had shown that female medical students have more positive attitudes toward learning communication skills than male medical stu-

dents[15,18,19,20]. While studies by Marambeetal and Shankaretal found no significant difference between male and female students in their positive attitude scores towards learning communication skills[13,21]. Study by Ullah stated that male students had significantly higher NAS scores than female undergraduate medical students[22]. Meta analysis had observed that female physicians were able to perform better in communication behaviors like collaborative, empathic communication and giving psychological information[18]. In our study PAS was higher in females and statistically significant as compared to males. Female students have better CS because of ease of giving information and partnership building.

In our study 92% students felt the need to improve their communication skills. Similar findings regarding students attitudes towards communication skills and their preferences for formal training in communications skills during the clinical training[21,22].

Communication skills in clinical practice are deemed as an essential component as they are instrumental in accurate diagnosis as well as convincing the patient to follow the given medical advice[14,23]. There had been studies where students' skills and confidence in communicating with patients had increased after training. Study by Yedidia concluded that educational communications model significantly improved third year students overall communications competence, skills in relationship building and also clinical competence[24].

Students appearing for United States Medical Licensing Examination (USMLE) are required to demonstrate competence in not only clinical but also interpersonal communication skills[4]. The General Medical Council (GMC) of the UK has encouraged that by the end of medical undergraduate program; medical students should have demonstrated and acquired proficiency in communication[20]. Medical schools in Malaysia have incorporated communication skills teaching sessions into the undergraduate medical curriculum for more than 10 years[25].

Effective communication can be learned during the academic years and practiced, this gives physician the confidence to transmit information to patients thereby improving patient compliance and finally helping to build a healthy doctor patient relationship[24]. CS learning programs definitely help in developing the communication skills in undergraduate and postgraduate students and this is eventually reflected in their clinical practice. CS is indispensable for medical practice and it can be taught and learned. Internationally, CS training has become an integral feature of undergraduate medical curricula[1,3,11].

In the twenty first century, teaching communication skills in undergraduate medical curriculum and training is considered to be of utmost importance[7]. The aim of erstwhile Medical Council of India (MCI) to standardize an Indian Medical Graduate for which MCI has realized the timely need of an effective Attitude, Ethics and Communication Skills (AETCOM) program for imparting communication skills as mentioned in its competency based undergraduate medical curriculum for the Indian medical undergraduate[8,26].

Symptom management and skilled communication with patients and families has become necessity in the coronavirus 2019 pandemic. It is important that clinicians responding to the coronavirus disease 2019 crisis need to be equipped with communication skills to support patient management and family communication. For patients and families, care from a clinician who communicates their caring could mitigate a lot of misunderstanding in communication between patient and COVID-19[27,28].

Study by Zayapragassarazan has stated the need of continuous and comprehensive internal assessment in assessing the core areas of attitude and communication[29]. Thus, in order to have a trustworthy doctor patient relationship and to provide comprehensive medical care, a formal training and evaluation of communication skills and its assessment needs to be followed in the current curricula[30]. This study gives a baseline glimpse of the positive and negative attitudes about communication skills. This will help to formulate and modify the communication skills module to improve communication skills among undergraduate medical students.

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